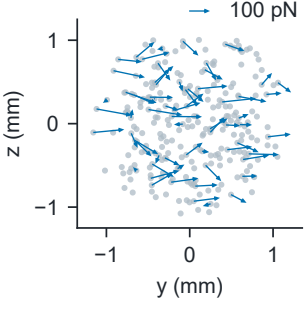
c Microbubble acoustic forces

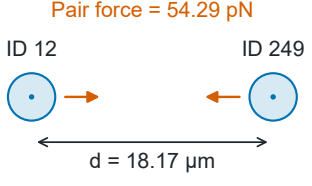
Cycle-averaged; t = 0

Primary Bjerknes force



$$\mathbf{F}_{P,i} = -\tfrac{1}{2} Re[\hat{V}_{i,1} \nabla P_i^{*}]$$

Secondary Bjerknes force



$R_0 = 1.78$ μm; true pair geometry; t = 0

$$\mathbf{F}_{S,ij} = \frac{\rho \langle \dot{V}_i \dot{V}_j \rangle}{4\pi d_{ij}^2} \mathbf{e}_{ij}$$